

Keshav Rajput

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Quantitative researcher / quant developer, new grad. MS CS, NYU Tandon; B.Tech CS and Applied Mathematics, IIIT Delhi.

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EDUCATION

New York University, Tandon School of Engineering

Master of Science (MS) in Computer Science

Sep 2024 - May 2026

GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi

Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics

2020 - 2024

CGPA 8.19/10

Coursework: Stochastic Processes, Statistical Inference, Convex Optimization, Game Theory, Econometrics, Valuation and Portfolio Management, Machine Learning

PROJECTS

trading-agent (not public) | Python, pandas, LightGBM, TabPFN, Chronos-2

2026

- Cost-aware backtesting: purged walk-forward with a 5-day embargo, CPCV, Deflated Sharpe, probability of backtest overfitting (0.06) and a random-portfolio null; 14 models on 8 US ETFs, 2011 to 2024.
- No candidate beat vol-targeted equal weight after costs (best 0.68 Sharpe vs 0.81), so none was promoted; zero-shot Kronos as a standalone signal called 5-day direction right 46.6%.
- Paper-trading runner with risk controls (10% target volatility, 35% per-ETF cap, exposure halved at 5% drawdown, kill switch at 10% drawdown or 3% daily loss); four gates before real money.

Non-stationary bandits | Python, NumPy, Matplotlib

2026

github.com/happyc0der/multi_armed_bandit_algorithms

- Five piecewise-stationary bandit algorithms (TS-GE, AdSwitch, M-UCB, UCB1, epsilon-greedy) behind one interface, benchmarked from 2 to 128 arms.
- Audited TS-GE against its paper and found the Beta update reversed: TS-phase regret 21,860 to 174; mandatory probing is 75.9% of its total regret.
- Made AdSwitch 59x faster (218.6 s to 3.7 s) and cut memory from 2,048 MB to 272 MB at K=128, T=1e6, bit-identical output.

Econometrics re-analysis (team of 5) | R, fixest, data.table, Python, pandas

2026

github.com/utkar22/Econometric-Analysis-of-Agriculture-and-Health-Across-India

2023 coursework with four teammates in Utkarsh Arora's repo; the 2026 re-analysis, pipeline, models, review and CI are mine.

- Re-analysed a 2023 econometrics team project on 673 Indian districts: reproduced both regression tables to 6 decimals, then showed the crop indices were row-order artefacts.
- Python ETL builds a 3,585-row district-year panel from Census 2011, NFHS-4 and IMD data; R (fixest) panel regressions with fixed effects and state-clustered standard errors show the headline effects vanish, within-R2 below 0.02.
- Wrote the review, fixed 7 run-breaking bugs in the archived scripts, added GPL-3.0 and CI that reruns the deterministic pipeline and fails on any changed table.

Kaggle S6E9, EV purchases | Python, LightGBM

2026

github.com/happyc0der/kaggle-s6e9-ev-purchases

- Kaggle Playground S6E9 (ROC AUC): out-of-fold 0.94648, public leaderboard 0.94649 against a leader at 0.94674; final ranking pending.
- Feature engineering with nested target encodings over the synthetic generator's artifacts on 668,665 rows, 20-fold cross-validation; logit blend of LightGBM, CatBoost and a 15-seed MLP.
- 192 experiments judged by paired DeLong hypothesis tests; label resampling showed a perfect model scores 0.9459 +/- 0.0003, so the gap left is label noise.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952. Final ranking pending, competition ends 2026-09-29.
- **TrafficFlowBench**, 2026. Public leaderboard 0.845 against a 0.553 baseline, with LightGBM time-series queue forecasting and an NNLS origin-destination solver. In progress, ends 2026-11-06.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Quant: backtesting (purged walk-forward, CPCV, Deflated Sharpe, PBO), econometrics (fixest, fixed effects, clustered SEs), time-series forecasting, bandits, Markov chains

ML: PyTorch, deep learning, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, NumPy, pandas, polars

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

LEADERSHIP AND ACTIVITIES

- Events head, Finnexia (finance club), IIIT Delhi; organised "The Critical Bid"